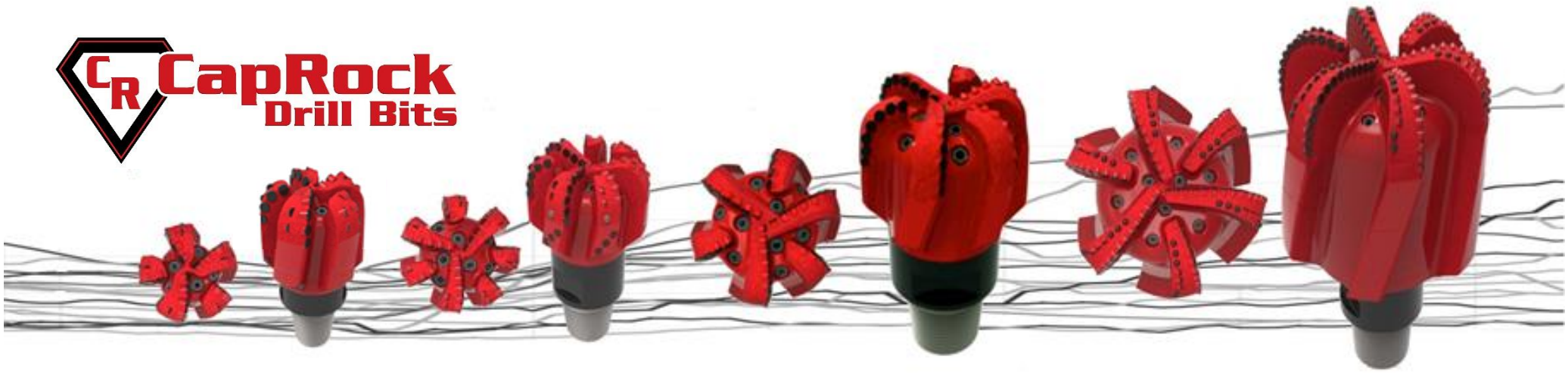
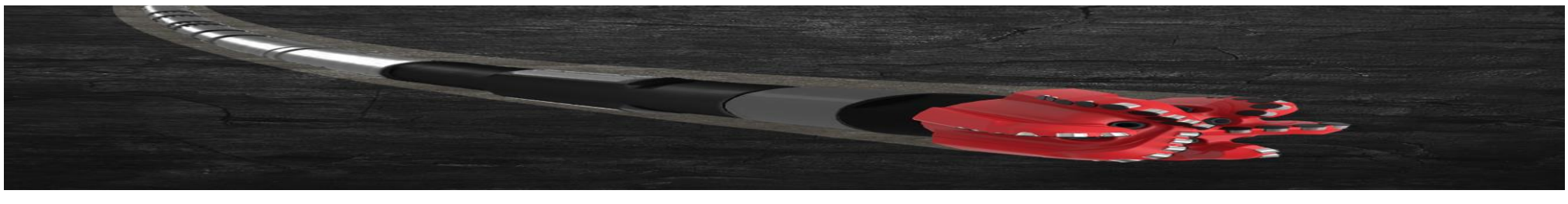
CapRock Drill Bits



Lithological Tops Callon & Cactus 157	Formation Remarks & DOCC Control Drilling Recommendations Casselman 40 #10UL	Drilling Parameters Midland County, TX	
Rustler (1,880') Salt/Anhydrite/Dolomite UCS: 10 – 15 kpsi	•Maintain stable WOB and RPM. Increase WOB up to 45K as needed after BHA is buried.	RPM Total 185-200	WOB (klbs) 30-45
Yates (2,980') Shale/Dolomite/Red Sands UCS: 15 – 20 kpsi	•Fast Drilling - Maintain a stable WOB and RPM.	RPM Total 185-200	WOB (klbs) 30-45
Queen (4,105') Dolomite/Sandstone UCS: 20-22 kpsi	•Consistent lithology •Monitor transition into the Grayburg as rock strength increases, low WOB can induce impact damage. Increase WOB in this section to keep cutters buried.	RPM Total 185-200	WOB (klbs) 35-45
Grayburg (4,550') Dolomite/Anhydrate UCS: 20 – 25 kpsi	•@**4,450'** Implement DOCC Transition Parameters. •Set total RPM at 170 and Control Drill @ 85ft/hr. •@**4,650**' Release DOCC Transition Parameters and resume with 180 RPM and 40-55K WOB.	RPM Total DOCC Set @ 170 Non-DOCC @ 180	WOB (klbs) DOCC Min = N/A DOCC Max @65K Other = 40-55
San Andres (5,175') Dolomite/Limestone UCS: 20 – 25 kpsi	•@**5,075'** Implement DOCC Transition Parameters. •Set total RPM at 170 and Control Drill @ 85ft/hr. •@**5,275**' Release DOCC Transition Parameters and resume with 180 RPM and 40-55K WOB.	RPM Total DOCC Set @ 170 Non-DOCC @ 180	WOB (klbs) DOCC Min = N/A DOCC Max @65K Other = 40-55
Clearfork (7,105') Conglomerate/Sandstone/Shale UCS: 27 kpsi	•@**7,005**' Implement DOCC Transition Parameters. •Set total RPM at 170 and Control Drill @ 85ft/hr. •@**7,205**' Release DOCC Transition Parameters and resume with 180 RPM and 40-55K WOB.	RPM Total DOCC Set @ 170 Non-DOCC @ 180	WOB (klbs) 40-55
Spraberry SS (8,110') Sandstone/Limestone UCS: 28 -32kpsi	•@**8,010**' Implement DOCC Transition Parameters. •Set total RPM at 170 and Control Drill @ 85ft/hr. •@**8,210**' Release DOCC Transition Parameters and resume with 170 RPM and 45-60K WOB.	RPM Total DOCC Set @ 170 Non-DOCC @ 170	WOB (klbs) DOCC Min = N/A DOCC Max @65K Other = 45-60
Sprayberry Section to TD *Approx. (8,600')*	•Maintain 170-180 RPM and Stable WOB to TD	RPM Total 170-180	WOB (klbs) 45-60

- **NOTES**
- **On all formations, except DOCC transition points, remove ROP set points on Auto Driller or set to 800FPH. In these non controlled intervals, use WOB set points.**
- **In all DOCC Transitions enable the ROP set point on auto driller. Remove WOB set points through transition zones or set max of 65K.**
- **Set max differential set point on Auto Driller to max differential pressure from motor spec sheet. Keep this set point for entire intermediate section.**